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Chapter 2
Artificial Intelligence Strategies for Cleaner and Greener Manufacturing

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ABSTRACT

The journal aims to reduce energy usage that emits greenhouse gases by making products more productive and eco-friendlier. The writing deals with the issue of how the current trends in Industry 4.0 and AI systems can be used to protect the environment from the harms of pollution. It incorporates 1) conserving energy, implementing eco-friendlier workplaces, and reducing carbon footprints and 2) digital transformation with environmentally positive implications. Also, like in real manufacturing, it addresses the possibility of AI-driven manufacturing. It refers to Industry 5.0 as an evolutional road of communication and people and machines collaboration that transforms both society and individual; personalization as a result and higher productivity are just a few foundations to transform our perception of production and everyday life. Other Firstly, our guide will give some examples of preventive measures. This part stresses investing in R&D, collaborating, complying, and ensuring decision-making is informed by data running through the supply chain.

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1. INTRODUCTION

This journal first questionnaire was eye-opening, because journal got information about AI techniques for clean and green production. Through this journal own personal research, it looked into the possible role between Artificial Intelligence and biodegradable packaging as a replacement to plastics. A practical option that has come to light is the utilization of starch amino as packing materials that offers a better option compared to traditional plastics.

AI has a number of applications that contribute to the optimum levels of biodegradable packaging materials usage as well. AI is capable of processing data related to the production, supply chain management logistics, and customer demand, which consequently, makes processes more streamlined and reduces the amount of waste. Such combinations of AI technologies can result in better utilization of available resources and can have a substantial impact in the reduction of harmful manufacturing practices.

Apart from this, measuring the success and effectivity of such measures also entails a period of monitoring and analysing results. Numbers like garbage recycling, carbon footprint reduction, and customer affinity to biodegradable packaging are useful indicators of how an artificial intelligence (AI)- based sustainability initiative is doing.

In the consideration of AI strategies as plausible sustainability improvements of biodegradable packaging, it should be stressed on by emphasizing the examples of successful implementation and best practices. Showcasing case studies that highlight the successful results of AI driven optimization solutions and the environmental benefits, is an effective way of making it clear for businesses.

In the end, AI strategies that target both the development of biodegradable packaging, economics and the possibility to scale up will get the first priority. Technology puts AI in manufacture with sustainability goals in mind, balancing out both sides thereby triggering positive environmental while at the same time meeting market demands.

2. PUBLIC SURVEY

The data of the AI application for cleaner and greener manufacturing was collected using a structured way in which different stakeholders representing varying perspectives and interests would be engaged. undefined

1. Survey Design: Overall, the first step was to build a detailed survey questionnaire that encompassed the main features of AI strategies, biodegradable packaging,

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sustainable production, and opinions of the stakeholders regarding those themes. The survey instrument was carefully crafted in order to be understandable, precise and to the point as well as relevant so that the respondents will provide meaningful answers.

2. Target Audience: The choice of respondents comprised of people from both the public and private sectors who were involved in the manufacturing industry, including experts, researchers, academics, and policymakers. The mission was to collect a wide variety of views and authority related AI and Sustainability in manufacturing sector.
3. Distribution Channels: Numerous routes to market including the use of traditional and nontraditional media were employed to reach the audience effectively. These channels included digital platforms, industry meetings, professional groups, and direct applications to pertinent groups and agencies (Rithik et al, 2024). The research was presented in an email campaign, social media posts and joint collaborations with industry partners to increase the rate of visitor as much as possible (Kathirvel, A., and A. K. Naren, 2024e).
4. Data Collection: The online platforms were used in conducting a survey to make reaching and participation easier. Each participant has been provided with a link to the survey questionnaire and they are free to complete it on their own time. The questionnaire combined multiple choices, open-ended, and rating scales so as to collect data both objectively and subjectively.
5. Data Analysis: Which was the next step of processing the data from the responses in a systematic manner (Kathirvel A, and Maheswaran C P, 2023). I use qualitative data inclusive of demographic information and responses frequency to analyze the data using data statistical tools (Subramaniam et al, 2020). Advanced collection of the qualitative data collected through open-ended questions was analyzed thematically in order to reveal the recurring patterns, insights and themes expressed by the respondents.
6. Validation and Quality Assurance: In order to verify the accuracy and the reliability of the data, procedures such as data validation checks, anonymity assurances, and random sampling method was applied. As part of data quality checks, there were processes in place to ensure any mistakes or faults in the responses were identified and corrected.
7. Reporting: To conclude, this step was the publication of the results of the survey along with the insights gathered at the end as summary report. The report highlighted decisive points, forecasting's trends, guidelines for green AI manufacturing, and the importance of the upcoming research and practice.

The data collection procedures used in the survey was designed to ensure that the process was transparent and inclusive whilst creating a robust and factual construct

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which leveraged diverse viewpoints on AI strategies for more Cleaner and Greener Manufacturing.

3. THEORETICAL FRAMEWORK

Climate Change as a result of a transformation on the world environmental level has emerged as one of the most scale problems of advancement and economic development. Increasing of greenhouse gas levels with high intensity of the crude carbon dioxide (CO2) emissions is an appropriate nexus of the electricity production and consumption, which is slowly but surely represents great hindrances for the environment. (Climate Monitoring, 2020). International Zones established to fight the climate problems in some cases shifted the commercial activities related to these public good to other countries. (Aukker et al., 2013)

However, this has led to an insignificant transfer of emissions in preference to a proper reduction (Gobinath et al, 2024). This trouble worsens when manufacturing is relocated to countries that rely upon electricity sources with high carbon emissions or use technology that might be more power-efficient (Hamela and Dr.A.Kathirvel, 2019a).

In this case, developing international locations face an unequal proportion of obligation, contributing significantly to 63% of greenhouse gas emissions and enduring the bulk of the negative results of climate exchange. The economic impact of climate change is especially significant in these areas, currently standing at 78% and projected to rise to 87% by 2035. (Nordhaus, 2017). The susceptibility of those residing in tropical nations to natural calamities and their inferior living conditions exacerbate their difficulties. In poor countries, adopting traditional methods to reduce emissions, such as implementing pricing mechanisms or switching to cleaner energy sources, may need to be revised due to the impracticality of abandoning fossil fuels or pursuing nuclear energy in the short term. (Geissdoerfer, 2017)

Given these difficulties, the idea of a circular economy presents itself as a practical and viable alternative, at least in the foreseeable future (Hamela and Dr.A.Kathirvel,, 2019b).. The concepts of circular economy, which aim to decrease emissions without significantly modifying energy infrastructure, are perfectly compatible with intelligent technologies. As a fundamental component of the product supply system and a significant contributor to greenhouse gas emissions, manufacturing plays a crucial role in the fight against climate change. Incorporating intelligent technologies into manufacturing processes is essential for attaining a more environmentally friendly and efficient paradigm.

The transition from “Industry 4.0” to “Industry 5.0” indicates a strategic shift inside the industrial paradigm, highlighting the significance of planetary limitations

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and societal objectives in production and economic advancement. The notion of "Intelligence" has served as a forerunner to this shift, emphasizing the importance of capturing and monitoring manufacturing operations to guide them toward environmental sustainability (Sendler, 2017,44). Industry 5.0 presents a new method focusing on providing long-term service for ecological and societal advantages, such as prioritizing human needs, promoting sustainability, and enhancing resilience.

Given the circumstances, transforming innovative manufacturing systems into a more environmentally friendly model during the industry 5.0 era is reasonable and crucial. Implementing intelligent technologies, known as "Intelligent manufacturing (GIM)," is a new and integrated manufacturing approach that facilitates the achievement of green objectives. (Deif, 2011). This research suggests that intelligent technologies can achieve different green and sustainable goals. It emphasizes the importance of continuously evaluating and improving the sustainability of these systems using green metrics. In the face of climate change, combining innovative technologies and eco-friendly production offers a hopeful path toward sustainable and ecologically aware industrial progress. (Atlas & Florida, 1998)

Currently, the emphasis on safeguarding the environment has grown increasingly vital. As societies globally acknowledge the pressing necessity to tackle environmental concerns, the notion of Green Smart Manufacturing (GSM) has evolved as a crucial contributor to this international endeavour. (Heller & Shukla, 2003) Since the industrial sector plays a significant role in economic advancement and energy usage, incorporating artificial intelligence (AI) into Green Manufacturing projects has become crucial to promoting sustainability. (Breque, 2021)

Governments actively promote the transition to Green Smart production because they recognize the importance of environmentally conscious production. (Zhou et al., 2019). The German Industry 4.0, Remaking America, and Made in China 2025 initiatives demonstrate a shared dedication to promoting sustainability in industrial methods. The recent (Kathirvel et al, 2024 a & b) upsurge in government backing has significantly increased industry and research on green manufacturing. (Duan et al., 2024)

Integrating artificial intelligence into Green Manufacturing processes, namely through the deployment of conversational and expressive robots, has significant promise. AI enhances industrial performance and aligns with the Sustainable Development Goals (SDGs), facilitating sustainable growth. By utilizing AI technologies, industries may make significant and purposeful contributions to the global environmental agenda.

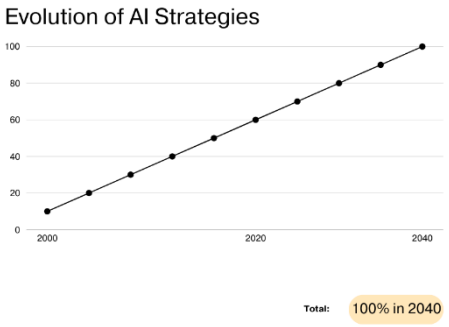
Incorporating Artificial Intelligence into Green Manufacturing is crucial in today's rapid technical progress and environmental consciousness. In the 21st century, combining technology and environmental preservation is essential for creating a sustainable and resilient future.

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The Development and Evolution of Greentelligent Engineering: Paving the Way for The Green Revolution in AI Strategies for Cleaner and Greener Manufacturing

The paper discusses the evolution and framework of Greentelligent Manufacturing, which aims to promote cleaner and greener manufacturing practices. It highlights the role of artificial intelligence strategies in driving the green revolution in manufacturing. (Sundar and Kathirvel, 2019).

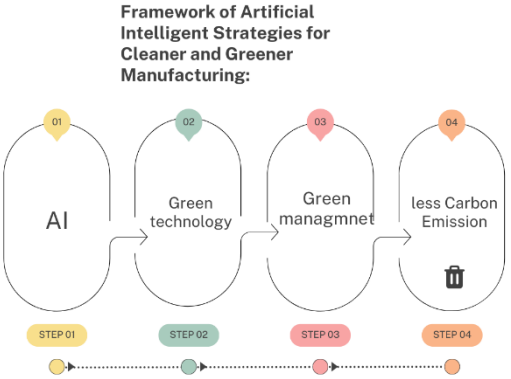
Figure 1. Evolution of AI strategies



Three degrees may be distinguished inside the evolution of intelligent manufacturing (GIM): emergent, transitional, and synergetic (Figure 1). The authentic desires of Intelligent Manufacturing (IM) and Green Manufacturing (GM) were sustainability and the exceptional use of available resources. GM and IM collaborated and pooled their talents to accomplish their objectives throughout the transitional segment. When GM and IM reached the synergetic degree, they laboured properly collectively, coordinating their goals and utilizing devices designed with each paradigm in thought. The evolution led to the development of intelligent manufacturing (GIM) as an intellectual manufacturing paradigm based upon humans and environmentally benign.

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Figure 2. Framework of artificial intelligent strategies for cleaner and greener manufacturing



"Greentelligent Manufacturing" aims to maximize efficiency, sustainability, and control in manufacturing processes by fusing green practices with intelligent technologies. The Greentelligent Manufacturing framework has been meticulously designed to encompass an extensive array of novel processes and materials, such as clean energy generation methods and the creation of eco-friendly products 'Intelligent' refers to incorporating contemporary sensor technologies, automation, and process optimization within the manufacturing environment, encompassing intelligent infrastructure, programs, and networked solutions (Kathirvel et al, 2020). In addition, it includes specialized "Information and Communication Technologies-ICT" that enable the real-time tracking of Waste, consumption, and greenhouse gas (GHG) emissions. It also facilitates the real-time control of energy, productivity, and costs across many sites. (Nimbalkar, 2017)

Integrating sensible technology into Green intelligent manufacturing is vital for achieving sustainable desires, staying aggressive, and enhancing the fee of advanced production industries. The framework ensures that business operations are green and environmentally responsible by integrating clever infrastructure and actual-time tracking.

Its objective is to achieve a more environmentally friendly approach by modifying industrial processes and reimagining product designs. Additionally, it prioritizes using ecologically friendly practices in manufacturing, such as decreasing emissions, optimizing resource utilization, and improving energy efficiency. Finally, it evaluates the environmental sustainability of the intelligent methods used in the production system. (Kumar & Rao, 2018)

In the following sections, we will examine Greentelligent Manufacturing more extensively, investigating the mechanisms that enable it and the tools and processes

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that support it. As we progress through this transformation, incorporating advanced technologies and environmentally friendly approaches becomes crucial for cleaner, more sustainable manufacturing practices. It is a significant milestone in the pursuit of the green revolution.

4. MATERIAL AND METHODS

This journal had the pleasure to do public questionnaire to help me understand AI strategies oriented towards cleaner and greener manufacturing. The reactions revealed an array of opinions and ideas that not only brought into focus the possible role of AI in manufacturing sustainability but also its major game changer character in this regard.

A highly ranked finding from the study was the role of AI in managing predictive maintenance. Participants agreed that AI-based predictive analytics is of great importance in preventing the emergence of defects in operation beforehand. With early detection of such problems, manufacturers can prevent machine downtimes and minimize material wastage with the potential to gain better operating efficiency in line with the environmentally friendly manufacturing approach.

Not only did the survey provided the optimism but it also showed how AI could double up in promoting the circular economy in the manufacturing sector. AI applications in supply chain management related to the product lifecycle emerged as highly sought after, particularly design optimization, reduction in overproduction using predictive demand forecast and planning, and intelligent routing (Kathirvel, 2024e). These tactics, in their view, should play a big role in making environmental footprint of the industrial operations less bulky.

It is interesting that some of the respondents remarked of the ethical implications of the introduction of AI in manufacturing. They then raised issues about data privacy, algorithmic biases and the need for transparency ownership in AI driven decision making processes. They argued that addressing these specific concerns would pave way for trust in artificial intelligence technologies and proper use of them in powering sustainability goals (Jolin et al, 2025).

The research also found that AI adoption is facing some challenges in the manufacturing sector. Low understanding of AI capabilities, high initial costs of implementation, and skills required for this field are among problems mentioned by topics experts (Kathirvel, 2024f0. Meanwhile, a number of the participants expressed their confidence in the possibility of an alliance between the players of industry, academy, and governmental institutions whose aims include overcoming the barriers and stimulating the adopttion of AI which is aimed at green and clean manufacturing processes.

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Lastly, the public survey on AI strategies for manufacturing that can be both cleaner and greener showed that the residents of the community hold a wide array of opinions and viewpoints. AI goes a long way in boosting resourcing efficiency to predictive maintenance and circular economy frameworks and, therefore, presents tremendous potential in leading the way for manufacturing sustainability.

Solving ethical issues as well as removal of barriers will be an important factor in using the AI to drive and achieve better environmental sustainability with industries.

5. EMERGING STUDIES THAT SUPPORT THE AI STRATEGIES IN GREEN MANUFACTURING: MANUFACTURING: A JOURNEY TOWARDS GREENER INNOVATION

Chen, Jiang, Yang, and Gao (2023) reported that the Global Cleaner manufacturing Conference (GCPC) 2023 showcased notable progress in cleaner manufacturing methodologies. Within the field of Greentelligent Manufacturing, Life Cycle Assessment (LCA) plays a crucial role as a standardized method for assessing environmental implications. Automation and artificial intelligence improve the efficiency of LCA by streamlining cost evaluations and connecting them to specific production processes for future projections. Advanced Information and Communication Technologies (ICTs) and the Internet of Things (IoT) allow for the continuous monitoring of CO2 emissions, which supports implementing a pricing model that considers the environmental impact. Furthermore, carbon taxes and Emissions Trading Systems (ETS) incentivize environmentally friendly behaviours, influencing producers and consumers worldwide. (Wara, 2007)

"Carbon Capture and Storage (CCS)" is a promising solution for achieving carbon neutrality but has obstacles. Norway is at the forefront of implementing "Carbon Capture and Storage (CCS)", viewing it as a significant endeavour comparable to a "Moon mission" in reducing carbon emissions, according to the authors Pérez-Fortes et al. in 2014. Intelligent technologies offer answers, which include virtual twins that verify strength usage and IoT tracking for optimizing plant operations. As we move towards sustainable production, these tools are essential for optimizing processes, minimizing emissions, and promoting a more environmentally friendly future.

Reconfigurable Manufacturing: Adapting for Tomorrow

Reconfigurable manufacturing (RM) demonstrates its ability to adapt and effectively handle wastewater and emissions by employing systematic reconfiguration. Integrating IoT technologies and AI in RM systems improves their capabilities

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by enabling early detection of machine faults, minimizing defects, and promoting efficient and sustainable manufacturing processes.

6. SIGNIFICANCE OF IMPLEMENTATION OF THE AI GREEN REVOLUTION

In today's increasing globalization and industrialization age, ensuring sustainability has become a top priority for enterprises. The expanded consciousness regarding environmental, social, and economic difficulties requires the implementation of ethical and sustainable business practices throughout supply chains. Sustainability, as defined by the World Commission on Environment and Development, refers to a type of development that meets the current generation's requirements while also considering the ability of future generations to fulfil their own needs. This definition recognizes the constraints imposed by limited natural resources (WCED, n.d).

To successfully navigate the intricate terrain of sustainability, it is necessary to tackle the social, economic, and environmental aspects, each of which has its own set of obstacles and changing importance as time progresses (Andrews, 2015). Attaining the entire sustainability of an employer poses a challenging task, requiring a complete approach that encompasses remodelling the commercial enterprise. This reorganization prioritizes the incorporation of Industry 4.0 (IDY four.0) technologies, "sustainable cleaner production", and "circular financial system-CE" practices.

Industry four.0 technology, which encompasses artificial intelligence techniques for more excellent environmentally friendly manufacturing, is becoming a critical element in this change. Artificial intelligence (AI), capable of making choices primarily based on data and automated approaches, offers organizations an effective device to improve their sustainability projects (Kathirvel et al, 2025). AI-enabled cleaner manufacturing techniques can reduce environmental effects and resource use. The round financial system, facilitated by utilizing superior technologies, promotes the most significant usage of resources through recycling and remanufacturing processes.

Figure 3. Waste management cycle



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AI techniques are becoming increasingly crucial for cleaner and greener production because they contain the answers to sustainability challenges in the best form. Incorporating Industry 4.0 technology is an excellent approach to increasing operational efficiency, reducing environmental impact, and promoting a circular economy. Users experience a complex picture of reality, and it is no longer only technological development but also an imperative strategic necessity as it is aligned with environmental sustainability and social welfare. AI programs are a tipping point for corporations, be it as gift essentials or as a way of ensuring that the future generation can do as well and not be left out of the equation. (Call for Papers - Journal of Cleaner Production | ScienceDirect.com by Elsevier, n.d.)

7. ADOPTION OF AI AND ITS IMPACT ON SUSTAINABLE PRODUCTION

Adopting AI innovations is a revolutionary force that propels manufacturers into an era of greater competitiveness, resilience, and environmental responsibility in pursuing sustainable manufacturing. AI strategies are characterized by the adoption of modular, automated processes that are guided by data. It allows for specific management of supply networks and fabric flows. The numerous benefits include more suitable competition, improved overall performance, and greater profitability. This wave of innovation may want to disrupt the production panorama by transforming production and transport techniques. It can also create a CO2 gasoline-free atmosphere and sell environmentally friendly production techniques.

Smart Factory technology, a defining characteristic of AI strategies, carries robotics, machine-to-system communication, and immediate decision-making into manufacturing operations. The final result is a more advantageous manufacturing cycle, ending in a faster product era, even optimizing capital expenditures. Incorporating synthetic intelligence (AI) techniques in Industry 4.0 enhances those benefits, facilitating the transition towards more environmentally friendly production methods. Data democratization and insight's introduction are prioritized to allow a seamless link past the flower production limits. It guarantees that sustainability concerns are incorporated into each part of the production environment. (Müller et al., 2018)

AI strategies necessitate a transformational industrial policy to create a sustainable environment. This approach should incorporate ongoing retraining and upgrading, spotting the dynamic nature of generation and automation. Establishing comprehensive commercial coverage for AI strategies is crucial to bringing together digital breakthroughs in different markets and areas. This policy alternate requires people to increase new capabilities that prioritize flexibility, the capacity to apply actual-time facts efficiently, and the potential to evolve line setups to fulfil daily

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requirements and adjustments. The emergence of Industry four.0 represents a vast transformation, necessitating a reevaluation of traditional factors of industries, consisting of production traces and supply chains, driven through novel technical possibilities and facts assets. (Habib & Chimsom, 2019)

The "Internet of Things-IoT" is critical in driving "Industry 4.0" with AI strategies projects by facilitating gathering statistics from numerous gadgets and sensors. Using AI and systems to gain knowledge of gear with records evaluation bureaucracy inspires sustainable manufacturing. The concept of Industry 4.0 facilitates integrating machines and programs into gadgets, resulting in an unbroken cycle of operation and machine-gaining knowledge. Organizations adopting Industry four.0 technology, including the "Internet of Things-IoT", synthetic intelligence (AI), and system getting to know, have experienced accelerated profitability and advanced performance in their operations. (Ammar et al., 2021) "Industry 4.0" is applied to the manufacturing sector to promote sustainable practices.

AI strategies effortlessly integrate with the current focus on sustainable manufacturing, facilitating the incorporation of sustainable development principles into the industrial sector. This integration leads to increased environmental, social, and economic efficiency. Industry 4.0 is a concept that encompasses the integration of digitalization technologies, big data, Cyber-Physical Structures, virtual reality, and cloud computing. Industry 4.0 has significant promise in sustainable manufacturing to increase productivity, streamline processes, and improve overall corporate efficiency. Nevertheless, the production sector has challenges and risks requiring sustainable methods. (Piccarozzi et al., 2018)

Big data, analytics, and deep learning are becoming critical factors in sustainable manufacturing, providing enhanced communication and efficiency in daily operations. In the energy industry, a significant change is happening, with digital electronic devices replacing paper. This transition is leading to new methods of functioning, communicating, and working with clients. (Chong et al., 2018)

AI strategies go beyond automation and include implementing sustainable practices in intelligent production. This evolution impacts smart grids, green energy management, and distributed output, contributing to a comprehensive approach to sustainability. As hardware vendors and software development businesses collaborate to construct and integrate corporate applications, the emphasis turns to ensuring internal reliability and adopting environmentally sustainable methods. The process of digitization, driven by AI strategies, not only provides new and creative ways of doing business but also accelerates the generation of renewable energy and the implementation of environmentally friendly energy supply strategies.

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8. ANALYTICAL APPROACH

Integrating the principles pursuing sustainable development goals:

The idea of "Pollution Prevention Pays" (PPP) combined with "Life Cycle Thinking" (LCT) to contribute to poverty reduction, climate change and social inequality can be realized by adopting a holistic environmental approach. The pollutant prevention program (PPP) adopts the strategy of catching and removing pollution right at the starting phase when it is still easy to deal with. It highlights that if pollution can be avoided, it will benefit the environment and the economy. By applying pollution control measures, organizations can decrease the amount of trash created, minimize the environmental impact, and cut down on waste disposal and treatment expenses. The order in which the mentioned factors have been placed is in line with SDG 12 (Responsible et al.), as it provides the necessary impetus for resource efficiency, waste reduction and sustainable practices.

LCT, or Life Cycle Thinking, is not limited to just one portion of the product or process. It is a broader view of the life cycle of a product or process, which incorporates the extraction of raw materials, the production process, and the final disposal. It aims to understand the surroundings and make products more effective from the design phase.

The results provided by the authors Chen, Jiang, Yang, & Gao, 2023). at the Global Cleaner Production Conference (GCPC) in September 2023 underline the relevance of sustainable production techniques. The cooperation framework of the PPPs and LCTs with sustainable development demanded a holistic strategic plan, which covered coordination between sectors, governments, and NGOs as the first step. Such culture must be acknowledged and perpetuated while allocating financial resources for technology and method development to prevent pollution. It could be that undertaking detailed environmental assessments, using sustainable production methods and cleaning up using the ecosystem principles are some ways of mitigating environmental impact. Furthermore, companies may apply the tools for lifecycle assessment and sustainability certificates to determine and spread information about the environmental impact of their products or services.

Governments have a crucial role in stabilizing and controlling sustainable practices by laying down the necessary policy and regulatory frameworks. Governments can establish pollution control regulations, provide financial incentives for environmentally friendly investments and help businesses adopt these methods through green procurement policies. Furthermore, governments should promote knowledge exchange and capacity-building programs as part of the strategy to transfer knowledge and skills that will help introduce PPP and LCT concepts to more sectors.

Civil society establishments, which include non-governmental businesses (NGOs), academics, and community groups, are critical in raising awareness, advocating for,

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and concerning stakeholders in sustainable improvement. Civil society may create top trade and ensure duty for environmental consequences by promoting pollutant avoidance and life cycle thinking, lobbying for legislative changes, and actively interacting with companies and legislators.

Ultimately, including pollution prevention and existence cycle wondering in pursuing sustainable development goals affords a radical approach to tackling environmental issues and promoting economic growth and social development. Businesses also lessen their environmental impact, improve aid performance, and help sustainable development objectives by enforcing proactive pollutant manipulation measures and thinking about the whole life cycle of their services or products. Practical cooperation among enterprises, governments, and civil society is crucial in promoting sustainable practices and advancing the shift towards a more sustainable and resilient future.

9. RESULTS

Incorporating Artificial Intelligence (AI) has become a powerful catalyst in driving sustainable and eco-friendly production. Although AI holds the potential to transform processes, minimize inefficiencies, and address the issue of greenhouse gas emissions, its deployment is full of risks and problems. Mastering those vital elements is essential for the effective use of AI in achieving waste-unfastened and greenhouse gasoline-unfastened manufacturing. (Bag, 2022).

Given the heavy reliance of AI systems on records, ensuring the privacy and safety of touchy information is an extensive problem. Highbrow property theft is drastically risky in economic enterprises, where special techniques and designs are giant. AI systems will rely on vast datasets for learning, and if data is peculiarly used or tampered with, it could have dangerous implications. To minimize those security issues, adequate cybersecurity measures must be implemented and comply with statistical security laws.

Figuring out the ethical form of using AI for industrial purposes is becoming more demanding daily. We can increase social, economic, and environmental inequalities with biases in AI algorithms. It occurs through the evolution of the gaps. For example, greener production might resort to an algorithm that is skewed techniques or components, subsequently hindering the original aim of sustainability. Ethical and fair processes must be considered when using artificial intelligence to avoid negative consequences. (Garetti, M, 2012).

Many manufacturing facilities use old structures that want to be mainly built to support superior technology consisting of synthetic intelligence (AI). Incorporating AI into the present-day infrastructure provides a significant obstacle. Manufacturers

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may be discouraged from adopting AI for environmentally friendly operations due to compatibility issues, retrofitting costs, and manufacturing interruptions during the changeover.

Deploying AI technology necessitates a considerable initial expenditure on hardware, software programs, and training. Manufacturers may be hesitant because of the ambiguity surrounding the return on investment (ROI) and the length required to attain demonstrable blessings. The capability for economic losses and the requirement for a long-term long-term outlook is probably intimidating. Effective dissemination of capacity lengthy-term benefits and conducive economic climate cultivation is essential for selling the significant adoption of AI within the production sector.

Introducing new technology, especially in industries with established techniques, now and again requires assistance with alternate resistance. Obtaining the help of stakeholders, control, and employees for AI's advantages in promoting environmentally pleasant production can be met with doubt and hesitance. Implementing change management processes, engaging in effective verbal exchanges, and highlighting early achievements

It is crucial to overcome resistance and inspire a way of life that is open to innovation.

Although implementing AI in manufacturing aims to minimize the environmental footprint, it is essential to behave through environmental impact checks (EIA). It is crucial to guarantee that AI operations no longer lead to pollutants or beneficial resource depletion. In addition, manufacturers should not forget the proper disposal of AI-related technology when giving up its existence cycle and mitigating any capacity environmental effects.

Manufacturing organizations must be subject to strict ecological regulations, and AI solutions do not simplify things. Verifying the AI methods in line with the established environmental requirements and gaining regulatory approvals may be complex and time-consuming. AI-enabled models present specific compliance challenges to manufacturers that need to direct assets and legal expertise to examine these complexities thoroughly but still need to match adherence to rules and operations requirements.

AI intelligence can immensely assist in enhancing supply chain transparency and traceability, the demolition of which is of much concern as far as sustainability is concerned.Also, as environmental authorities consider implementing devices in a challenging global supply chain, they encounter several issues. The implementation strategy for the sustainability of the production process must include the routine check-up of all materials and elements to meet the required criteria. This is why the collaboration with providers and blockchain technology we can do to traceability is at its best.

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For AI systems to bring about lower manufacturing tactics and operating costs, manufacturing AI hardware often results in ecological expenditures. Challenges arise from the mining and processing uncommon elements for the AI era, electricity-intensive manufacturing tactics, and digital waste control. Achieving a harmonious equilibrium between the blessings of synthetic intelligence and the ecological effect of its hardware necessitates a complete and all-encompassing strategy in the direction of sustainability.

Integrating AI in business industries gives capacity advantages and problems as the purpose is to reap waste-free and greenhouse gas-loose operations. Thorough deliberation is required because of the hazards related to statistics privacy, moral concerns, the integration of outdated structures, and the environmental consequences of the manufacturing AI era. To surmount those problems, a collective endeavour must encompass prominent figures in the region, policymakers, and the labour pressure. By adopting a scientific and accountable method, synthetic intelligence (AI) can function as an influential associate in accomplishing environmentally friendly and green manufacturing methods, thereby facilitating the improvement of a sustainable business future.

10. STRATEGICAL RECOMMENDATION FROM RESULTS

Above are the challenges which can be associated with the following recommendations: By using artificial intelligence (AI) to achieve environmentally friendly and pollution-free manufacturing processes, it is possible to develop several strategic suggestions to tackle these crucial issues (Jovane et al.,2008)

Dedicate resources to R&D endeavours to create AI technologies customized for sustainable manufacturing. It includes financial programs promoting advancements in renewable energy, waste minimization, and carbon capture technology. Bhamra (2016)

Promote cooperation among industry players, government agencies, academic institutions, and technology suppliers to expedite the implementation of AI-powered solutions. Collaborative endeavours may facilitate the exchange of optimal methods, combine resources, and stimulate joint efforts to achieve sustainability objectives.

Keep updated with changing rules and standards for environmental protection and emissions reduction. Efficiently incorporate these criteria into industrial processes and allocate resources to technology that guarantees adherence to present and future laws.

Adopt a method of making decisions based on data using AI analytics to maximize the use of resources, reduce waste, and find ways to increase efficiency. Develop

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and deploy reliable data gathering and analysis systems to monitor environmental performance parameters and facilitate ongoing improvement efforts.

Allocate resources towards implementing training initiatives that enhance the skills and capabilities of workers, enabling them to run AI-enabled industrial systems proficiently. Establish and foster an organization's sustainability and environmental stewardship culture to align with its strategic goals.

Broaden the use of AI beyond internal industrial operations to include the entire supply chain. Engage with suppliers and logistics partners to deploy artificial intelligence-based solutions to manage inventories, optimize transportation, and implement sustainable sourcing practices.

Establish mechanisms for ongoing monitoring and assessment of environmental performance measures to measure progress towards sustainability objectives—Utilise real-time data analytics to pinpoint areas of improvement and adapt strategy appropriately.

Investigate potential investments in renewable energy sources, such as solar, wind, and hydroelectric power, to decrease dependence on fossil fuels and carbon emissions linked to industrial processes. (Arnold, Kiel, and Voigt 2016)

Manufacturers may become leaders in sustainable industrial practices by prioritizing these strategic guidelines, which will help them overcome the hurdles of applying AI for greener and cleaner production.

11. APPLICATIONS: FOR AI STRATEGIES FOR ACHIEVING CLEANER AND GREENER MANUFACTURING

AI techniques significantly improve within the commercial zone, wherein physical sports and virtual generation are integrated, resulting in a transformative impact. AI techniques usher in a new generation of optimal performance, approaches, and development in production through mixing devices, gaining knowledge of and real-time facts processing. This advancement also contributes to a cleaner and more environmentally friendly production technique. This paradigm shift enhances productiveness, efficiency, and flexibility and promotes sustainability, leading to extended profitability and toughness for companies. (Sarmiento et al., 2020)

Factories Which are Sustainable and Clever

One of the number-one ways to introduce Intry 4.0 is to convert production facilities into environmentally pleasant and intelligent factories. Companies are reorganizing their production procedures to beautify their environmental friendliness and sustainability. It includes integrating interconnected production gadgets, energy

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structures empowered with synthetic intelligence, and gadgets for the industrial Internet of Things. Implementing AI technologies helps with accountable intake and improvement by effectively dealing with natural resources through circular financial system concepts and weather cognizance.

Minimized Carbon Emissions and Pollution Management

AI techniques are essential in mitigating carbon emissions and managing pollutants in many industries. These sustainability dreams are achieved with robotics, renewable energy, statistics collection, IoT, and digitalization. Industry 4.0 complements development in fields that include solar power consumption, energy safety, and lightweight materials by reducing the reliance on conventional fossil fuels and optimizing aid usage. (Dincer and Acar, 2015)

Eco-Friendly Structures

Industry 4.0 has many applications in creating sustainable buildings, from architecture to continuing management. Architects utilize Building Information Modeling (BIM) software to enhance the sustainability of buildings. Digital technologies facilitate the use of eco-friendly bacteria that possess the inherent ability to repair structures, thereby demonstrating the potential of sustainable design and architecture. Intelligent systems actively control heating and cooling to optimize energy usage in real time, improving overall environmental effectiveness.

Utilizing Digital Information for Beneficial Environmental Influence

Industry 4.0 incorporates digital information and communication technology to address and reduce the favourable environmental effects caused by different businesses. Analyzing manufacturing processes and quantifying the positive consequences of intelligent manufacturing are achieved through digital technologies. It entails a thorough digital transformation endeavour to improve operational efficiency, workplace health and safety, and collaboration with product design departments.

Increased Sustainable Advantages

AI strategies and technologies provide a comprehensive solution for dealing with the intricacies of technology, manufacturing, and sustainability. The advantages of sustainability are improved by assessing and grouping production procedures according to enabling technologies such as 5G cellular wireless connectivity, IoT,

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machine learning, and augmented reality. Practical applications arise, enhancing operational procedures, environmental friendliness, and collaboration between manufacturers, suppliers, and customers.

Service Management

AI strategies achieve efficient service management by connecting measurement devices with manufacturing businesses' comprehensive information and automation infrastructure in energy and services management. It entails gathering, transmitting, and retaining immense energy supply utilization, generation, and shift data. The widespread availability of data and improved accessibility to computing resources (Kathirvel et al, 2024c) enable the utilization of advanced artificial intelligence methods for identifying variables and analyzing patterns. (Waibel et al., 2017)

Enhancement of Productivity

One of the main objectives of Industry 4.0 is to enhance operations' efficiency, competitiveness, and overall financial commitment. Industry 4.0 enhances proactive resource use in enterprises by prioritizing energy management. Different power generation sources must be integrated into a market that is both increasingly demanding and spread out to maximize energy efficiency (Kathirvel, 2024d).

Improving Environmental Innovation

Industry 4.0 encompasses environmental advancements through optimizing automated manufacturing processes, leading to improved efficiency and less waste generation. The Internet of Things (IoT) is a network that links machines, sensors, actuators, and computers together to gather and exchange information. Digital twins are virtual representations of natural properties and systems that offer valuable insights for future planning, data collection, and device surveillance. Cyber-physical systems combine computational, network, physical processes, and environmental advancements within the system (Chin et al.,2019)

Enhancing Production Efficiency

Cloud computing networks work together to supply data to optimize production, predict process failures, and reduce downtime. Interoperability facilitates the effective exchange of knowledge among various entities such as units, sensors, actuators, devices, robots, and human beings. Big data and analytics pertain to the extensive and ever-changing datasets obtained from diverse sources of machine

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learning technology. Industry 4.0 is a field that focuses on developing strategies, ideas, and solutions to optimize safety and production in industries. It, in particular, emphasizes the use of prevention techniques. (Nahavandi,2019)

Enhance Proficiency and Adaptability in Automation

Training and practice structures are essential in equipping people with the competencies to thrive in developing Industry 4.0 professions. The past few centuries' commercial enterprise and industrial sector advancements encompass various technological components associated with Industry 4.0. Robots are essential for implementing bendy automation and offer numerous benefits, such as extra excellent manipulation, a more secure printing environment, reduced bottlenecks, improved performance, and enhanced worker pride. Robotic integration can improve environmentally sustainable strategies

Eco-Friendly

Elevated degrees of automation enhance manufacturing and worker protection. Heavy machinery may additionally be managed using far-flung control and haptic remarks technology over 5G networks.

Tackling Climate Change

AI strategies offer practical techniques and instruments to tackle climate change and other sustainability goals, leading to favourable industry outcomes. Partnerships among technological pioneers, scientists, private enterprises, and non-profit entities play a crucial role in the swift progress of the worldwide economy. The advent of digital technology brings forth novel consumer prospects and transforms business practices, thereby impacting the long-term sustainability of economies. This technology can significantly influence the protection, inclusivity, affordability, and long-term growth of all those involved.

Manage Entire Industrial Operations

Industries that aim to automate their industrial operations recognize the importance of access to current and relevant data. Connecting to cloud storage may ensure all pertinent data is accessible anytime and anywhere. Industry 4.0 technology, such as cameras and sensors, offer supplementary data to assist engineers in achieving jobs with more efficiency. Integrating nanotechnology with 3D printing enhances production by increasing flexibility and improving energy efficiency.

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Reduce Waste

The automotive sector wholeheartedly adopts lean methods, prioritizing eliminating waste in all manufacturing parts. Adopting lean thinking allows manufacturers to reduce resource consumption, including packaging, raw materials, and energy. The response to climate change presents an opportunity to increase revenue, expand industries, and cultivate a sustainable image. Implementing intelligent manufacturing techniques can effectively reduce waste and minimize the emission of greenhouse gases while enhancing production processes.

12. IMPLEMENTING INNOVATIVE SUPPLY CHAIN MANAGEMENT PRACTICES TO ACHIEVE SUSTAINABILITY GOALS IN THE ERA OF CARBON TRADING

In the midst of the developing worry for natural maintainability and the pressing need to lessen fossil fuel byproducts, production network the executives (SCM) systems have acquired huge significance in achieving manageability targets, particularly in carbon exchanging. In the midst of the gigantic extension of the worldwide economy, fossil fuel byproducts have turned into a significant block to accomplishing economical turn of events. Producing firms, which assume a critical part in fossil fuel byproducts, face developing requests to carry out low-carbon rehearses all through their stockpile chains. Moreover, there is a developing accentuation on carrying out reusing and remanufacturing answers for items that have arrived at the finish of their life cycle. It is finished to help the shift towards a round economy, which plans to expand asset utilization and save energy.

To address these difficulties, creative supply chain management strategies must be used to promote carbon reduction and improve sustainability outcomes. Carbon trading systems, which facilitate the exchange of carbon credits between businesses, have become prominent as efficient instruments for controlling carbon emissions from industrial facilities. Governments are also regulating by creating regulation laws on carbon emissions monitoring to make carbon trading an integral part of worldwide initiatives on climate change management. So this is why a good supply chain management (SCM) strategy application should be aimed at the period of carbon trading in order to accomplish sustainability objectives and considerably reduce carbon emissions.

This special issue wants to have a research section that includes all studies that are innovatively focused on designing supply chain management systemic methods that can meet the carbon trading era requirements. The discussions and encouragements provided in this study will develop exploitable concepts and techniques that

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facilitate trading costs and, thus, the sustainability achieved by concentrating on transformative advances in SCM.

Introducing emissions of carbon issues into the supply chain management process is a significant feature signifying what the focus is on. Conventionally, cost and productivity have led enterprises to develop supply chain strategies. However, organizations that trade carbon must deal with both sides of the coin and focus on making their procedures environmentally friendly and with fewer carbon emissions. Through the promulgation of carbon emissions data into a decision support system, businesses can now develop advanced supply chain management methods and make science-based choices. The former makes reusing and recycling the primary components, increasing the product lifecycle and reducing the environmental load.

Furthermore, ensuring that the materials are purchased sustainably is essential for promoting carbon offset within the supply chains. Organizations may greatly reduce emissions by obtaining them independently or through suppliers with the least carbon consequence. The latest SCM (Supply chain management) is technology-driven and employs data analytics that utilizes information on suppliers' eco-friendliness to find issues that need cooperation and enhancements. By forming long-term alliances with suppliers who constantly push for environmental sustainability, the concept of the supply value chain can be boosted, encouraging greener production chains and global carbon footprint reduction.

We also focus on applying eco-friendly logistical and transport management systems. As transportation, to a considerable extent, enables CO2 emissions within the logistics chain, completing its processes efficiently that considers the environmental impact is the key to minimizing the negative effects. The eco-friendly SCM approach focuses on improving carrier routes, adopting high-tech vehicle types, and promoting other ways of transportation, such as rail and marine transport. Besides that, shifting to electric vehicles and using alternative renewable energies for logistics activities result in significant carbon target achievement.

13. FUTURE SCOPE

Within the manufacturing industry, the future is poised to provide several prospects and breakthroughs due to the incorporation of AI techniques to achieve cleaner and more environmentally friendly production. A vital development place is the ongoing improvement and broadening of Industry 4.0 technology. With the growing recognition of this technology, we may assume a shift in production techniques towards improved sustainability and environmental friendliness.

Industry 4.0 has established the basis for greater sensible and practical manufacturing by incorporating automation, records analytics, and device getting-to-know.

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The destiny improvement toward Industry 5.0, defined by progressed cooperation and communique among humans and machines, has exceptional capacity. This trade will adjust the interplay between human beings and machines, creating a more collaborative and flexible manufacturing environment.

In addition, the capacity for simulated intelligence in assembling goes past the capacity's substantial space and comprises of the whole organization of suppliers and business frameworks (Sundar and Kathirvel, 2020a). Through digitization and availability, makers can improve asset use effectiveness, decline waste, and diminishing nursery fuel outflows over the creation cycle. Taking on this far-reaching way to deal with manageability is fundamental for handling critical natural issues and accomplishing carbon impartiality goals.

Furthermore, incorporating AI-powered technology will empower manufacturers to get the right of entry to higher productivity, efficiency, and creativity stages. AI technologies can beautify competitiveness and pressure corporate growth by enabling predictive renovation, satisfactory control, custom-designed product creation, and actual-time decision-making (Sundar and Kathirvel, 2020b).

AI is fast becoming a reality so that producers can take over the current bottlenecks, and the future provides a playground for digital manufacturers. Incorporating AI technology to improve the manufacturing techniques of the business realm can be instrumental in creating a more sustainable and responsible business zone that will ultimately help the betterment of the forthcoming generations.

14. CONCLUSION

Journals survey on the employment of AI tools for Cleaner and Green Manufacturing has yielded useful information and views on the role AI can play in the process of providing sustainability inside manufacturing. Journal have utilized a quantitative method of gathering data through a structured survey methodology which has helped us understand the stakeholder perspectives, trends and opportunities in this area.

The level of attention and understanding on the part of industry players on the applications of AI in driving cleaner and greener manufacturing activities is high with the business sector. The feedback given encouraged awareness in the growing concern of the negative role traditional manufacturing processes play and the need for sustainable solutions.

The survey portrayed AI as a powerful tool for the optimization of resource utilization, predictive maintenance, and supply chain management as the leading themes. Citizen comments focused on the ability of AI to be used in algorithms to make data analyses for real-time changes, less waste utilization, and improved production efficiency, which contribute to greener manufacturing.

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While on one hand AI-oriented sustainability approaches hold a promise, they have their dark side being the high cost, skills necessary to use them and ethical issues among which. Nevertheless, it was stated that the joint initiatives involving industry partners, research institutions and governance were catalysts for overcoming these obstacles and to moving towards greener production processes.

A hypothesis can be developed from the survey results, which then would recommend some strategies to optimize AI systems in Cleaner and Greener Manufacturing. These are consisting of nurturing cross-sector partnerships, facilitating talent for AI, regulating frameworks that trigger responsible practice, and constantly using AI for progress and innovation.

As a whole, the survey illustrates the capabilities of AI as a powerful tool that enables a positive environmental output in the manufacturing industry. Bridging AI-driven strategies, using ecofriendly materials such as biodegradable packaging, and addressing the challenges through taking measures collectively, the manufacturers can build the foundation that the future will rely on to have its sustainability goal accomplished.

Nevertheless, regardless of those boundaries, the advantages of integrating AI into business operations outweigh the potential drawbacks. Multiple case studies and research effects have proven that AI-driven solutions notably beautify operational efficiency, product first-rate, and environmental sustainability. By using artificial intelligence (AI) to improve industrial procedures, companies may also enhance their competitive gain even by actively participating in international initiatives to deal with weather change and conserve herbal assets.

Nevertheless, the robust incorporation of AI into manufacturing requires collaborative endeavors from stakeholders in each public and business domain name. Governments need to offer beneficial guidelines and incentives to promote AI generation. Simultaneously, industry leaders should allocate resources closer to instructing and growing their personnel to facilitate an unbroken transition to AI-pushed commercial processes.

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